



*Mechanical
Engineering
Department*

**KONYA
TÜRKİYE**



ENGINEERING MECHANICS STATICS

| SIXTH EDITION |

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1

INTRODUCTION TO STATICS

1. Introduction to Statics

Mechanics

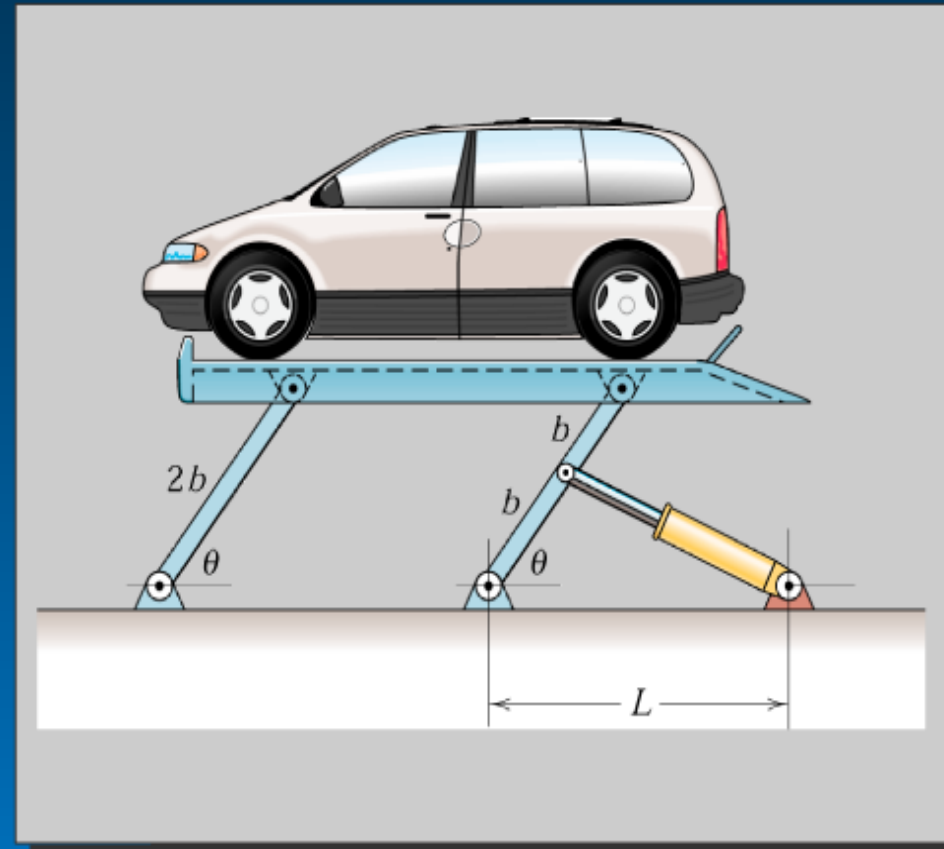
- a. Definitions
- b. Newton's Laws
- c. Units
- d. Law of Gravitation

Mathematics

- a. Review of Vectors
- b. Accuracy and Approximation

Problem Solving

- a. Guidelines



Definitions

Mechanics: The applied science that treats the state of rest or motion of **bodies** that are subjected to forces.

Statics: The branch of mechanics that treats objects which are stationary (usually) or at constant velocity.

Kinematics: The study of the relationship among displacement, velocity, and acceleration without regard to the cause of the motion.

Kinetics: The branch of mechanics that examines the relationships between forces and acceleration.

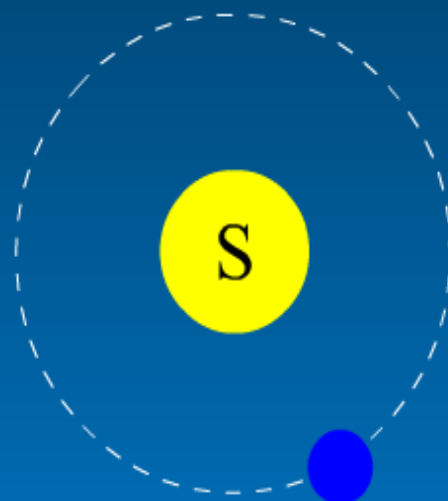
Dynamics: Combination of kinematics and kinetics.



Types of Bodies

Particle: Usually defined as a body with finite mass but not significant size; more practically, a body whose size is unimportant for the problem at hand.

Example: Earth in orbit about Sun.



Particles

Elastic Bodies

Rigid Bodies

Fluids

Dismiss

Types of Bodies

Elastic Solid: Under the action of a shear force F that does not cause the elastic limit to be exceeded, an elastic solid will go to the deformed configuration and stop. Removal of the force allows the body to return to its original shape.

Examples: Steel, rubber

Particles

Elastic Bodies

Rigid Bodies

Fluids



Apply shear force

Dismiss

Types of Bodies

Elastic Solid: Under the action of a shear force F that does not cause the elastic limit to be exceeded, an elastic solid will go to the deformed configuration and stop. Removal of the force allows the body to return to its original shape.

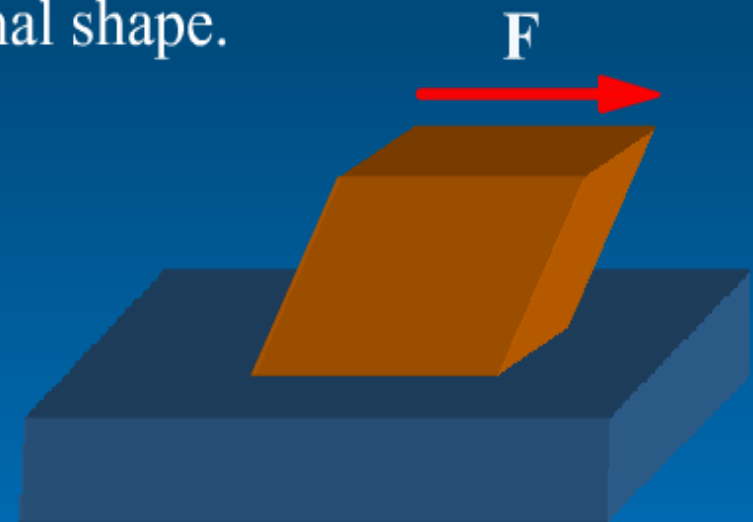
Examples: Steel, rubber

Particles

Elastic Bodies

Rigid Bodies

Fluids



Remove shear force

Dismiss

Types of Bodies

Rigid body: A body of finite size that does not deform, even under the action of large forces. If points A and B are arbitrarily chosen, the straight-line distance d will not change.

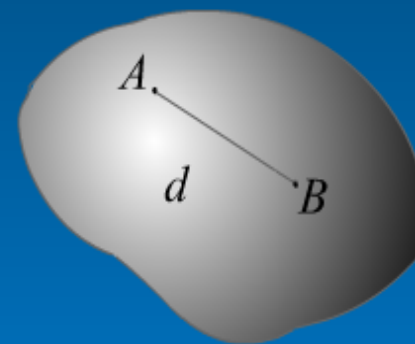
Examples: None exists! (But in many cases the deformation is so small that it may be neglected).

Particles

Elastic Bodies

Rigid Bodies

Fluids



Dismiss

Types of Bodies

Fluid: The body will continue to deform as long as the shear force is present.

Categories: Liquids will not fill a container uniformly, even in an orbital environment. Gases fill a container fairly uniformly.

Examples: Liquids: Tar, glass!; Gases: Air, Helium

Particles

Elastic Bodies

Rigid Bodies

Fluids

Dismiss



Definitions

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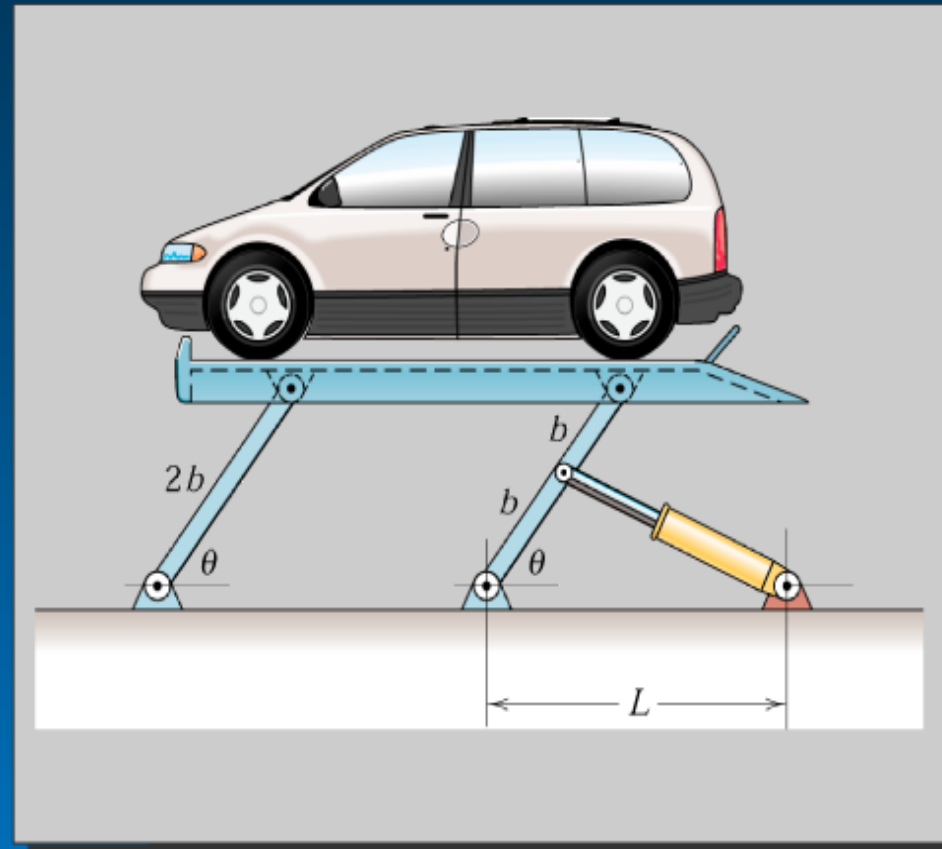
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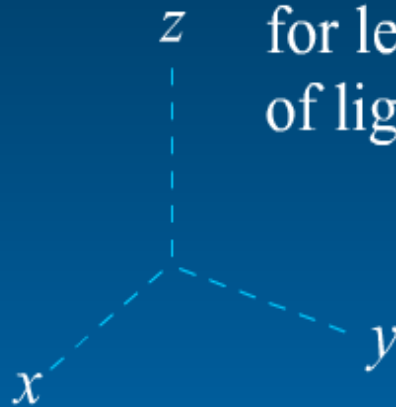
- a. Guidelines



Newton's Laws

Four fundamental qualities involved in Newton's laws:

Space: Three dimensional Cartesian region; the definition for length (foot or meter) is based upon a wavelength of light under specified conditions.

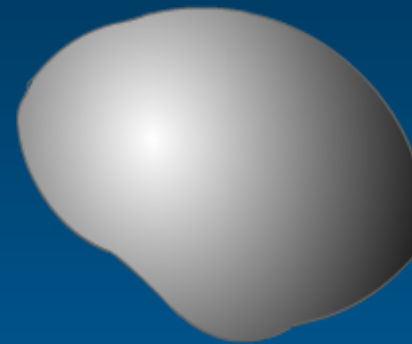


Time: The second is based upon a carefully specified atomic event.



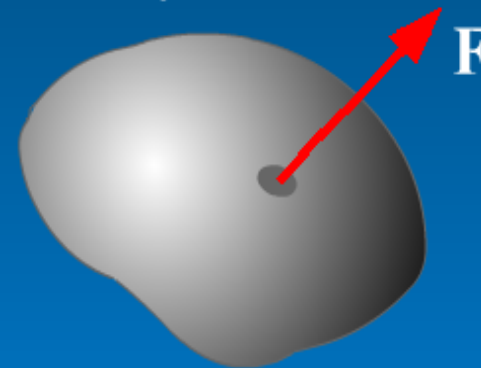
Four fundamental qualities involved in Newton's laws:

Mass: A measure of the resistance of a body to linear acceleration.



Force: A push or pull exerted on a body, characterized by

- * magnitude
- * direction
- * point of application



Law I

A particle remains at rest or continues to move in a straight line with a uniform velocity if there is no unbalanced force acting on it.

Law II

The acceleration of a particle is proportional to the resultant force acting on it and is in the direction of this force.

Law III

The forces of action and reaction between interactive bodies are equal in magnitude, opposite in direction, and colinear.

Equation statement of Law II:

$$\mathbf{F} = m\mathbf{a}$$



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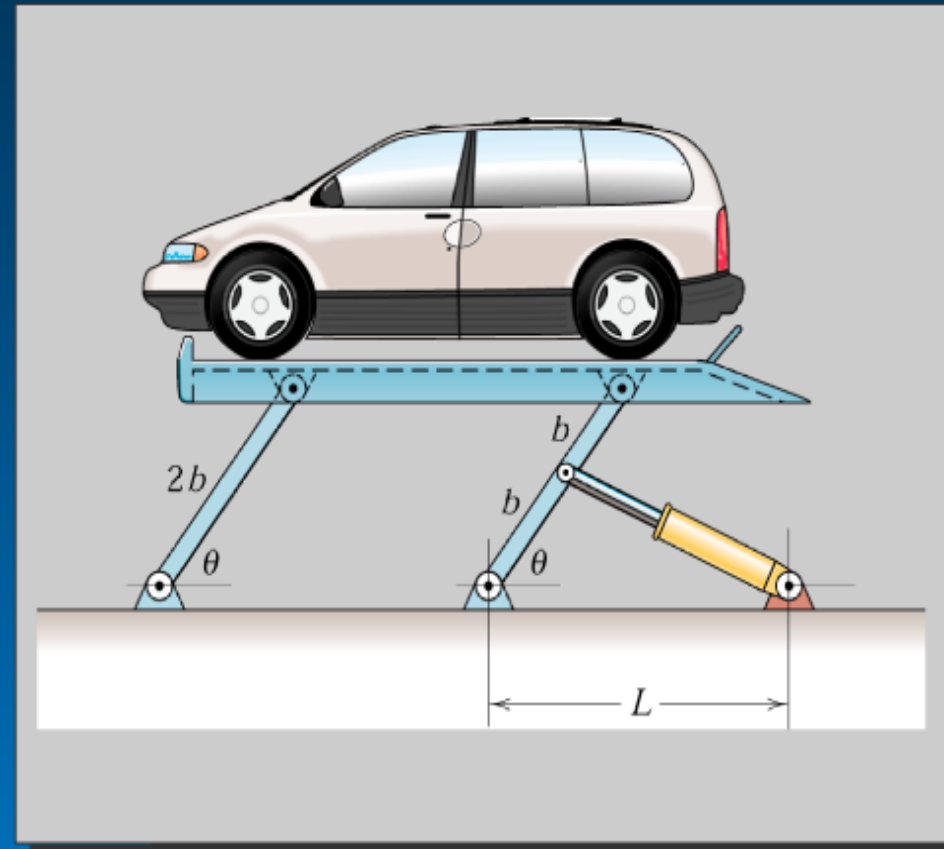
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Units

Units	SI	U.S.
Mass	kilogram (kg)	slug (-)*
Length	meter (m)	foot (ft)
Time	second (s)	second (sec)
Force	newton (N)*	pound (lb)

*not base or fundamental units, but defined by Newton's second law.

Note: The SI system is an absolute system, because a gravitational field is unnecessary for definition of the three basic units. On the other hand, the U.S. customary system is a gravitational system.



Kilogram (kg)

The mass of a standard platinum - iridium kept at the International Bureau of Weights and Measures near Paris, France.

Meter (m)

1 650 763.73 wavelengths of the orange-red radiation (light) emitted by the krypton - 86 atom.

Second (s)

The duration of 9 192 631 770 periods of oscillation of the light emitted by the cesium - 133 atom.

Dismiss



Foot (ft)

12 inches in one foot, where 1 inch = 2.54 centimeters (exactly).

Pound (lb)

The force of gravitational attraction (weight) acting on a standard mass under specific conditions.

Dismiss



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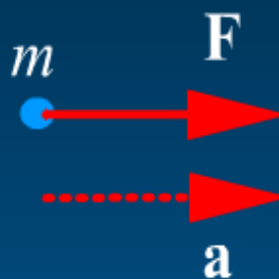
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(SI)

$$\mathbf{F} = m\mathbf{a}$$

$$1 \text{ N} = (1 \text{ kg}) (1 \text{ m/s}^2)$$

$$1 \text{ N} = 1 \text{ kg} \cdot \text{m/s}^2$$



(U.S.)

$$\mathbf{F} = m\mathbf{a}$$

$$1 \text{ lb} = (1 \text{ slug}) (1 \text{ ft/sec}^2)$$

$$1 \text{ slug} = (1 \text{ lb} \cdot \text{sec}^2) / \text{ft}$$

So:

- * 1 N is the force which will accelerate 1 kg at 1 m/s²
- * 1 slug is the mass which a 1-lb force will accelerate at 1 ft/sec²



2/2



1. Introduction to Statics

Mechanics

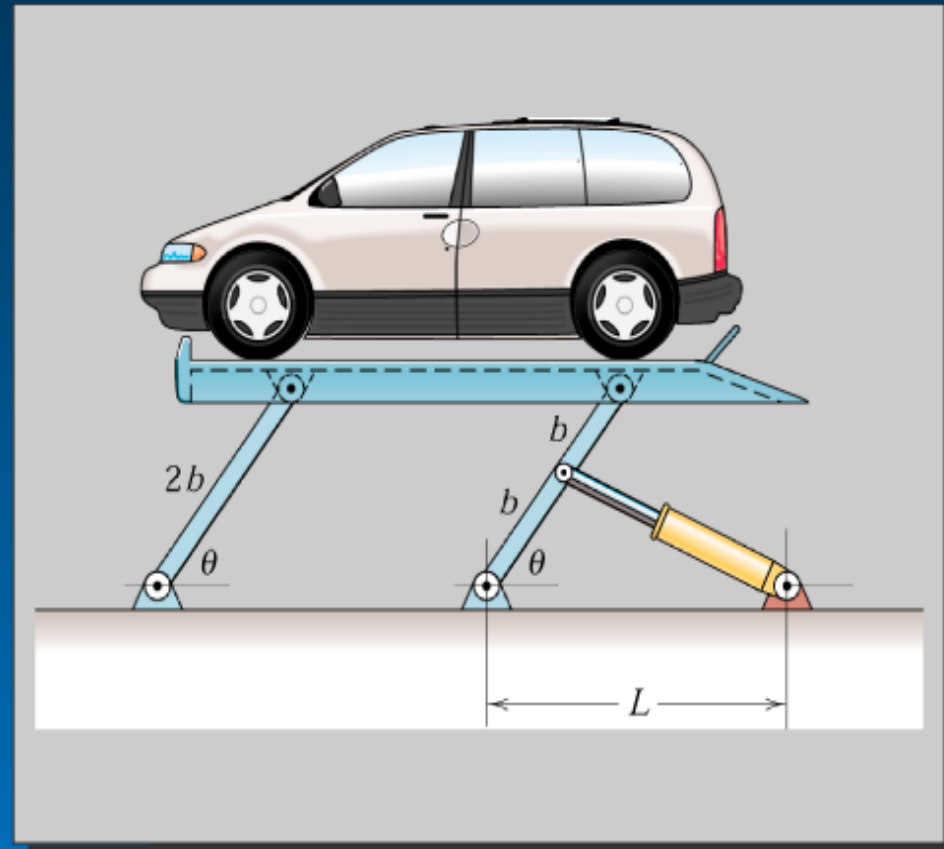
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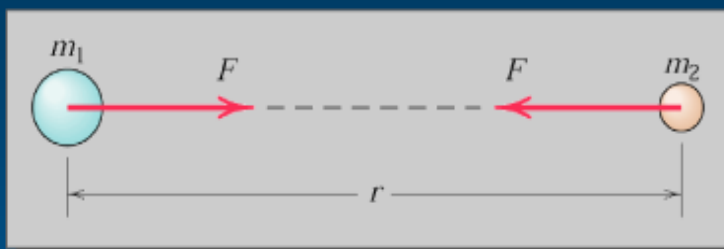
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Law of Gravitation



m_1, m_2 = the masses of the two particles

r = the center-to-center distance between the particles

The magnitude F of the mutual force of attraction is

$$F = G \frac{m_1 m_2}{r^2}$$

where G is the universal gravitational constant.



Special case: mass m on surface of Earth

From $F = G \frac{m_1 m_2}{r^2}$, we have

$$F = G \frac{m_e m}{R^2} = m \left(G \frac{m_e}{R^2} \right)$$

or $W = mg$

where W is the magnitude of the force of gravitational attraction (the weight of m) and $g = \frac{G m_e}{R^2}$ is the acceleration of gravity.

$$\text{SI: } g = \frac{6.673(10^{-11}) 5.976(10^{24})}{[6371(10^3)]^2} = 9.825 \text{ m/s}^2$$



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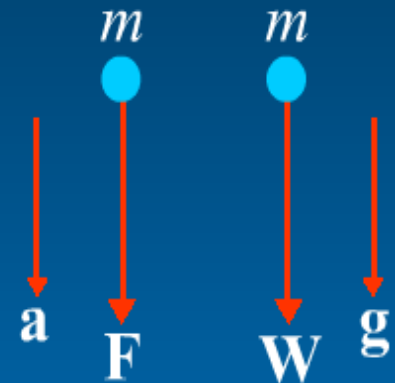
Special case: mass m on surface of Earth (cont.)

$$\text{U.S.: } g = \frac{3.439(10^{-8}) 4.095(10^{23})}{[3959 (5280)]^2} = 32.23 \text{ ft/sec}^2$$

These are absolute values of g relative to a *nonrotating* Earth. The commonly values for analyses relative to the *rotating* Earth are:

$$g = 9.81 \text{ m/s}^2$$

$$g = 32.2 \text{ ft/sec}^2$$



Note that $W = mg$ may be obtained from Newton's Second Law applied to a particle of mass m which is allowed to fall freely near the surface of the Earth.



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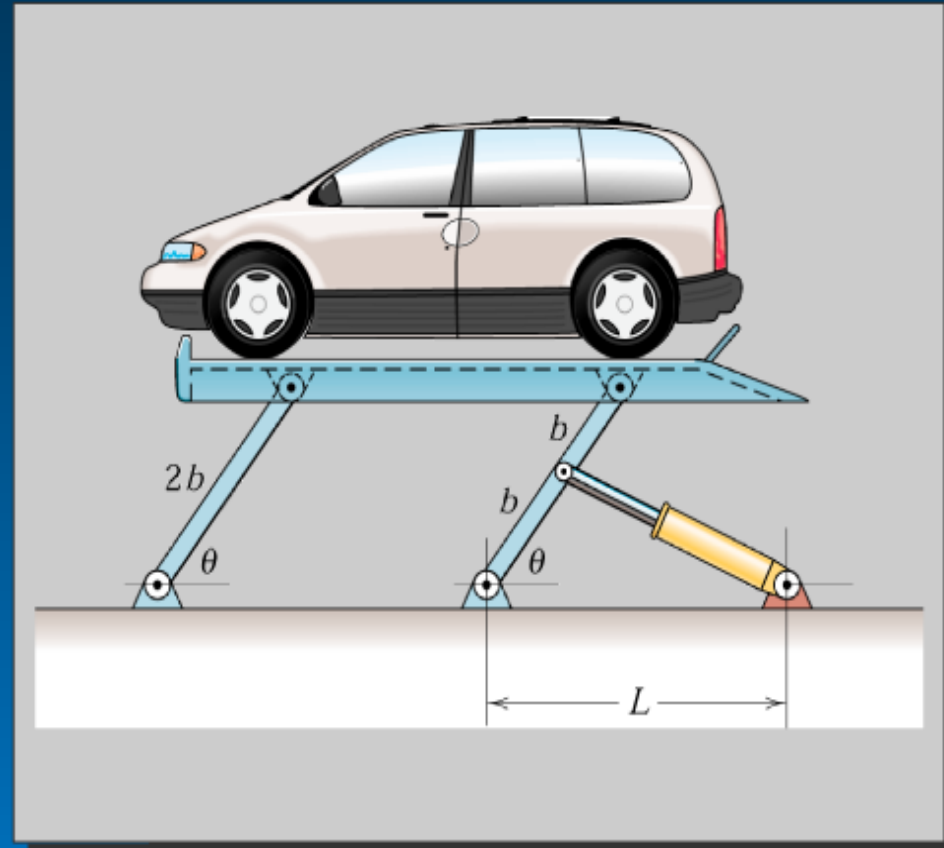
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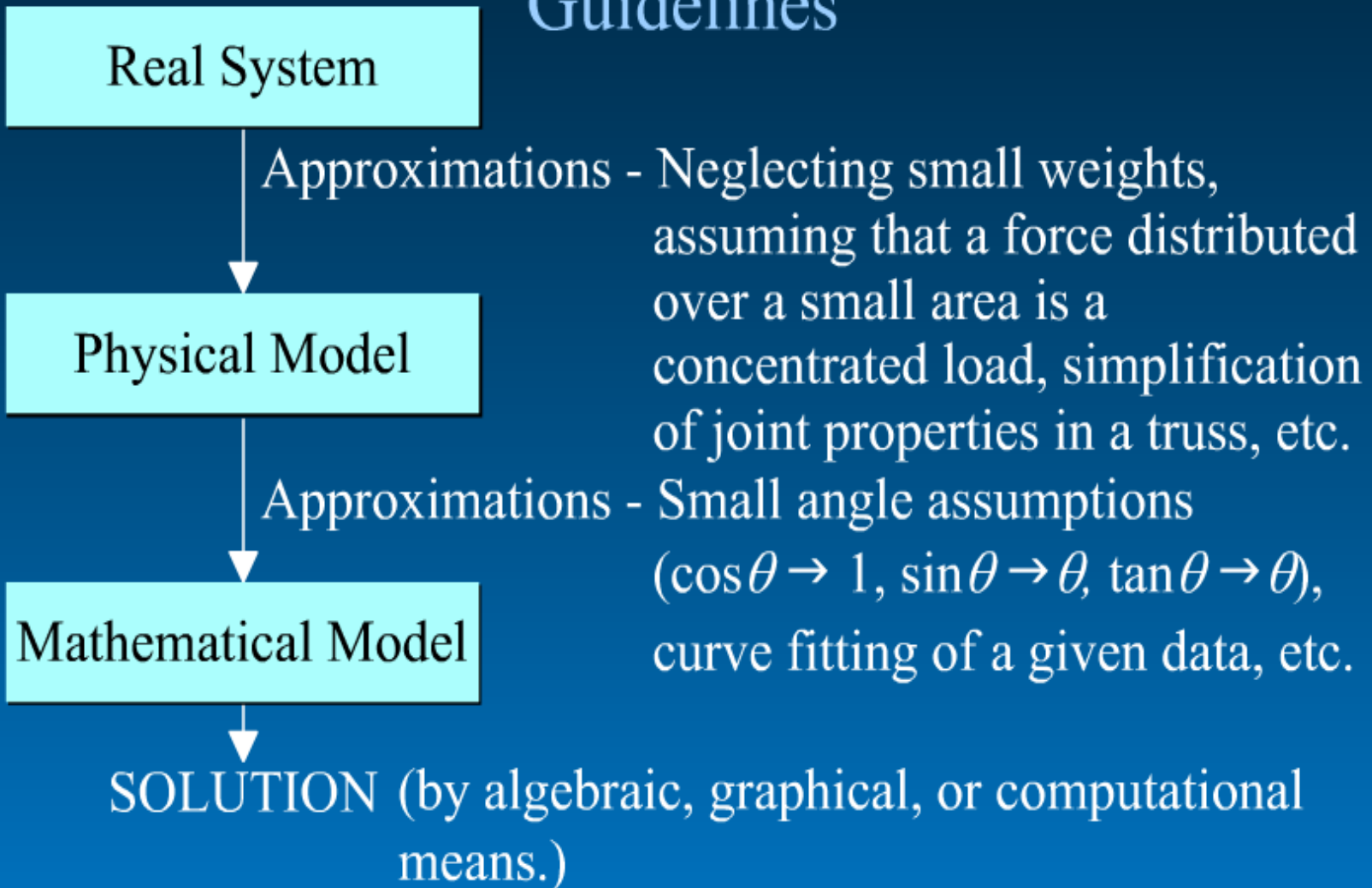
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Guidelines



Problem Presentation

1. Use 8 1/2" by 11" paper - fronts only, stapled in upper left, unfolded.
2. Use pencil (no pens) along with straightedge / templates for all diagrams.
3. Begin each problem on a new page (exception: very short problems).
4. Clearly identify vector quantities.



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Problem Presentation (cont.)

5. Format: Given:
 Find:
 Solution:

Clear statement of method(s) utilized

Indicate any coordinate system used

Identify answer(s) (box or double underlined)

6. Important: A complete free-body diagram (FBD) is required for all problems involving static equilibrium or kinetics.



Sample Problem

Determine the mass in slugs of a person whose weight is 200 lb. Convert the given weight to newtons and then determine the corresponding mass in kilograms.

Solution:

From $W = mg$, we have

$$m = \frac{W}{g} = \frac{200 \text{ lb}}{32.2 \text{ ft/sec}^2} = 6.21 \text{ slugs}$$

From the table of conversion factors inside the front cover of the book, we see that $1 \text{ lb} = 4.4482 \text{ N}$. Thus, the person's weight in newtons is

$$W = 200 \text{ lb} \left[\frac{4.4482 \text{ N}}{1 \text{ lb}} \right] = 890 \text{ N}$$

Finally, the mass in kilogram is

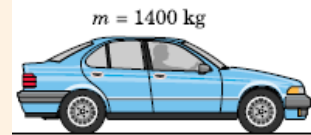
$$m = \frac{W}{g} = \frac{890 \text{ N}}{9.81 \text{ m/s}^2} = 90.7 \text{ kg}$$

As another note to the last result, we can convert slugs to kg. Again using the table inside the front cover of the textbook, we have

$$m = 6.21 \text{ slugs} \left[\frac{14.594 \text{ kg}}{\text{slug}} \right] = 90.6 \text{ kg}$$

The slight discrepancy is due to the fact that we used $g = 32.2 \text{ ft/sec}^2$ above rather than the more accurate $g = 32.174 \text{ ft/sec}^2$.

Sample Problem 1/1



Determine the weight in newtons of a car whose mass is 1400 kg. Convert the mass of the car to slugs and then determine its weight in pounds.

Solution. From relationship 1/3, we have

$$\textcircled{1} \quad W = mg = 1400(9.81) = 13\,730 \text{ N} \quad \text{Ans.}$$

From the table of conversion factors inside the front cover of the textbook, we see that 1 slug is equal to 14.594 kg. Thus, the mass of the car in slugs is

$$\textcircled{2} \quad m = 1400 \text{ kg} \left[\frac{1 \text{ slug}}{14.594 \text{ kg}} \right] = 95.9 \text{ slugs} \quad \text{Ans.}$$

Finally, its weight in pounds is

$$\textcircled{3} \quad W = mg = (95.9)(32.2) = 3090 \text{ lb} \quad \text{Ans.}$$

As another route to the last result, we can convert from kg to lbm. Again using the table inside the front cover, we have

$$m = 1400 \text{ kg} \left[\frac{1 \text{ lbm}}{0.45359 \text{ kg}} \right] = 3090 \text{ lbm}$$

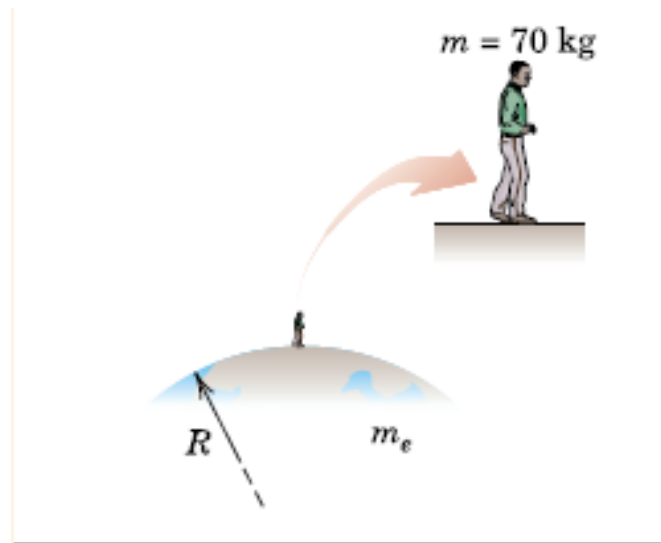
Sample Problem 1/2

Use Newton's law of universal gravitation to calculate the weight of a 70-kg person standing on the surface of the earth. Then repeat the calculation by using $W = mg$ and compare your two results. Use Table D/2 as needed.

Solution. The two results are

$$\textcircled{1} \quad W = \frac{Gm_em}{R^2} = \frac{(6.673 \cdot 10^{-11})(5.976 \cdot 10^{24})(70)}{[6371 \cdot 10^3]^2} = 688 \text{ N} \quad \text{Ans.}$$

$$W = mg = 70(9.81) = 687 \text{ N} \quad \text{Ans.}$$





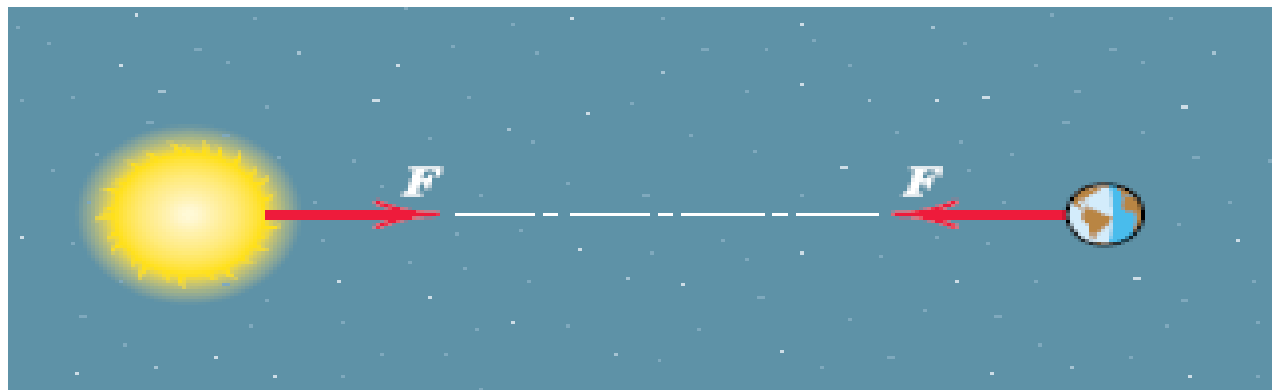
1/7 Determine the weight in newtons of a woman whose weight in pounds is 125. Also, find her mass in slugs and in kilograms. Determine your own weight in newtons.

Ans. $W = 556 \text{ N}$, $m = 3.88 \text{ slugs}$, $m = 56.7 \text{ kg}$

$$\boxed{1/7} \quad W = (125 \text{ lb}) \left(\frac{4.4482 \text{ N}}{1 \text{ lb}} \right) = \underline{556 \text{ N}}$$
$$m = \frac{W}{g} = \frac{125}{32.2} = \underline{3.88 \text{ slugs}}$$
$$m = \frac{W}{g} = \frac{556}{9.81} = \underline{56.7 \text{ kg}}$$

1/9 Compute the magnitude F of the force which the sun exerts on the earth. Perform the calculation first in newtons and then convert your result to pounds. Refer to Table D/2 for necessary physical quantities.

Ans. $F = 3.55(10^{22})$ N, $F = 7.97(10^{21})$ lb



1/9

$$F = \frac{G m_e m_s}{d^2} = \frac{3.439(10^{-8})(1)(333,000)(4.095 \cdot 10^{23})^2}{(92.96 \cdot 10^6 \cdot 5280)^2}$$

$$= \underline{7.97(10^{21}) \text{ lb}}$$

$$F = 7.97(10^{21}) \text{ lb} \left(\frac{4.4482 \text{ N}}{1 \text{ lb}} \right) = \underline{3.55(10^{22}) \text{ N}}$$



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